Machine Learning

From Intuition to Implementation



A textbook for BCA students
Course code NBCA-601

Unit I Introduction · Unit II Supervised Learning · Unit III Ensembles & SVM
Unit IV Unsupervised Learning · Unit V Neural Networks

How to Use This Book

Learning Objectives

Every chapter opens with a box like this one, telling you exactly what you will be able to do by the end.

Key Idea

The one idea to remember. If you remember nothing else, remember this box.

Definition: Term

Formal definitions you may be asked to write in exams.

Example

Worked examples with real numbers, solved step by step.

Common Mistake

Mistakes students commonly make, and how to avoid them.

Try It Yourself

Short Python experiments. Run them, change numbers, and watch what happens.

Exam Tip

Frequently asked exam questions and how to answer them.

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Chapter 1

Introduction to Machine Learning

Learning Objectives

After this chapter you will be able to:

- explain the difference between **AI, Machine Learning and Deep Learning**;
- define Machine Learning using Tom Mitchell's (T, P, E) definition;
- classify ML systems as **supervised, unsupervised, semi-supervised** or **reinforcement**;
- distinguish **batch vs. online** and **instance-based vs. model-based** learning;
- list and explain the main **challenges** of ML, especially **overfitting** and **underfitting**;
- train your first model in Python in under ten lines.

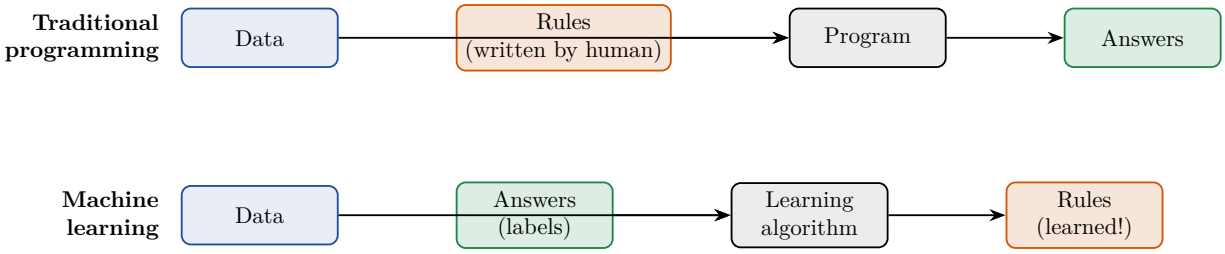
> *Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.*
>
> — Arthur Samuel, 1959

1.1 Why Do We Need Machine Learning?

Imagine you must write a program that detects *spam e-mail*. The traditional way is to write rules by hand:

- If the subject contains "FREE" ⇒ spam.
- If the sender is unknown and the mail contains "winner" ⇒ spam.
- If . . . (hundreds more rules)

Spammers quickly change their words ("FR33", "fr.ee") and your rule list becomes long, fragile and impossible to maintain. A **machine-learning** program instead looks at thousands of mails already labelled as spam or not-spam and *discovers the pattern by itself*. When spammers change tactics, we simply retrain it on new data.

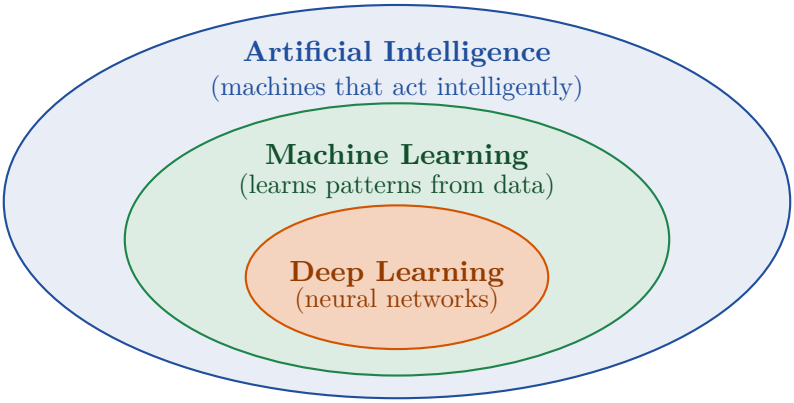


Key Idea

In traditional programming, humans supply *data + rules* and get *answers*. In machine learning, humans supply *data + answers* and the computer finds the *rules*.

1.2 Artificial Intelligence, Machine Learning and Deep Learning

These three terms are often mixed up. They are nested, like Russian dolls.



Definition: Artificial Intelligence

Artificial Intelligence (AI) is the science of building machines that perform tasks that normally require human intelligence, such as reasoning, understanding language, recognising images and making decisions.

Definition: Machine Learning

A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E. (Tom Mitchell, 1997)

Example: Spam filter in (T, P, E) form

Task T	Flag new e-mails as spam or not spam
Performance P	Percentage of e-mails classified correctly (accuracy)
Experience E	A collection of e-mails labelled by users

If accuracy rises as we feed in more labelled mail, the program is *learning*.

Exam Tip

"Define Machine Learning" appears almost every year. Write *both* Samuel's one-line definition and Mitchell's (T, P, E) definition, then give the spam-filter example. This earns full marks.

1.3 Where Is Machine Learning Used?

Area	Application	ML task
E-mail	Spam filtering	Classification
E-commerce	"Customers also bought"	Recommendation
Health	Detecting disease from X-rays	Image classification
Transport	Self-driving cars	Perception, control
Language	Google Translate, chatbots	Sequence modelling
Finance	Credit-card fraud detection	Anomaly detection
Real estate	Predicting house price	Regression

1.4 Types of Machine Learning Systems

We classify ML systems using three independent questions.

Q1. Is it trained with human supervision? (supervised, unsupervised, semi-supervised, reinforcement)

Q2. Can it learn on the fly? (batch vs. online)

Q3. Does it compare with known data or build a model? (instance-based vs. model-based)

1.4.1 Supervised Learning

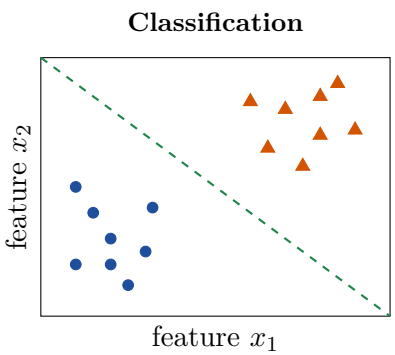
The training data includes the **answer** (called a *label* or *target*). It is like studying with an answer key.

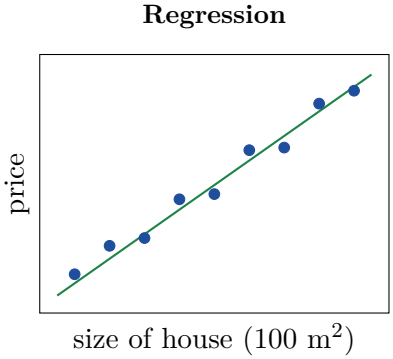
Definition: Supervised Learning

Given training examples $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n}$, where $\boldsymbol{x}_i$ is the input (features) and y_i the correct output (label), learn a function f such that $f(\boldsymbol{x}) \approx y$ on new, unseen inputs.

There are two main kinds of supervised task:

	Classification	Regression
Output y is	a *category*	a *number*
Example	spam / not spam	price of a house
Typical algorithms	k-NN, Naive Bayes, SVM	Linear regression





1.4.2 Unsupervised Learning

The training data is **unlabelled**; the system must find structure on its own, like sorting a pile of unknown coins into groups.

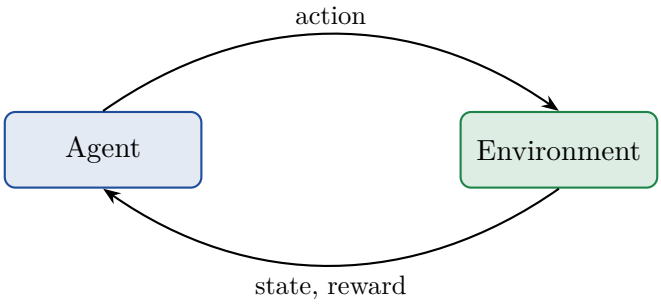
- **Clustering:** group similar customers (e.g. K-Means, DBSCAN) — Unit IV.
- **Dimensionality reduction:** compress many features into few (e.g. PCA) — Unit IV.
- **Anomaly detection:** spot unusual credit-card transactions.
- **Association rules:** "people who buy bread also buy butter".

1.4.3 Semi-supervised Learning

Labelling data is expensive, so we label *a little* and leave most unlabelled. Google Photos is a classic case: it clusters faces automatically (unsupervised) and you type a name once per person (supervised).

1.4.4 Reinforcement Learning

An **agent** observes an environment, takes **actions**, and receives **rewards** or penalties. It learns a *policy* that maximises total reward, as when a robot learns to walk or AlphaGo learns Go.



Type	Labels?	Feedback	Typical use
Supervised	Yes	Correct answer	Spam detection, price prediction
Unsupervised	No	None	Customer segmentation
Semi-supervised	Few	Partial	Photo tagging
Reinforcement	No	Reward / penalty	Games, robotics

1.4.5 Batch vs. Online Learning

Batch (offline): trained once on *all* the data, then deployed. To learn new data you retrain from scratch.

Online (incremental): learns continuously from small mini-batches as data arrives, so it suits stock prices or streaming data and needs less memory. Its *learning rate* controls how fast it forgets old data.

1.4.6 Instance-based vs. Model-based Learning

Instance-based: memorises the training examples and predicts for a new point by comparing it with similar ones (e.g. k-Nearest Neighbours).

Model-based: builds a mathematical model from the examples and then uses *only the model* to predict (e.g. linear regression).

Example: Does money make people happy?

Suppose we have data on GDP per capita x and life satisfaction y for several countries, and the points look roughly linear. A model-based approach assumes

$$y = \theta_0 + \theta_1 x$$

and *learns* the best parameters θ_0, θ_1 from the data. For a new country with GDP x_{new} we predict $\hat{y} = \theta_0 + \theta_1 x_{\text{new}}$ and never look at the training data again.

Try It Yourself

Your first model, using the `scikit-learn` library:

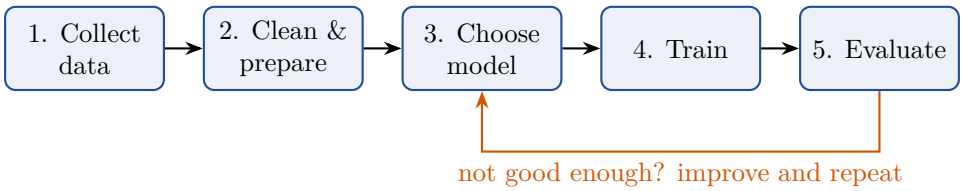
```
from sklearn.linear_model import LinearRegression

# Experience E: house size (100 m^2) and price (lakh Rs)
X = [[1], [2], [3], [4], [5]]
y = [15, 26, 29, 44, 46]

model = LinearRegression()          # choose the model
model.fit(X, y)                     # learn theta_0 and theta_1
print(model.intercept_, model.coef_[0])
print(model.predict([[6]]))         # price of a 600 m^2 house?
```

Challenge: add two more data points and see how the prediction changes.

1.5 The Machine Learning Workflow



1.6 Main Challenges of Machine Learning

"What can go wrong?" falls into two groups: *bad data* and *bad algorithm*.

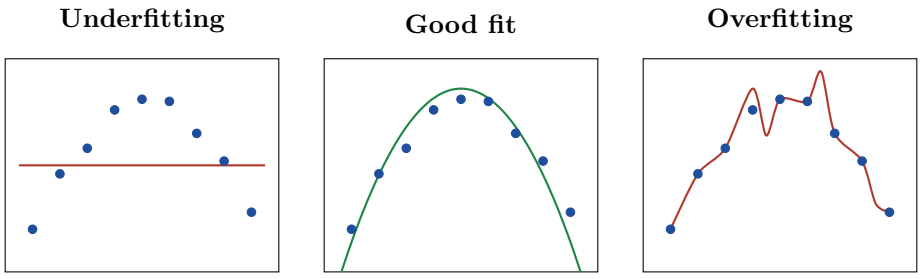
1.6.1 Problems with the data

1. **Insufficient training data.** Simple tasks need thousands of examples, and complex tasks like image recognition need millions. Banko & Brill showed that, given enough data, even simple algorithms perform almost as well as sophisticated ones.
2. **Non-representative data.** If you survey only rich countries, your model of "happiness vs. money" fails for poor ones. This is called **sampling bias**.
3. **Poor-quality data.** Errors, outliers and missing values make it hard for the system to see patterns, so data cleaning takes a large share of every ML project.
4. **Irrelevant features.** "Garbage in, garbage out." **Feature engineering** means selecting useful features, extracting new ones by combining old ones, and creating new ones by collecting more data.

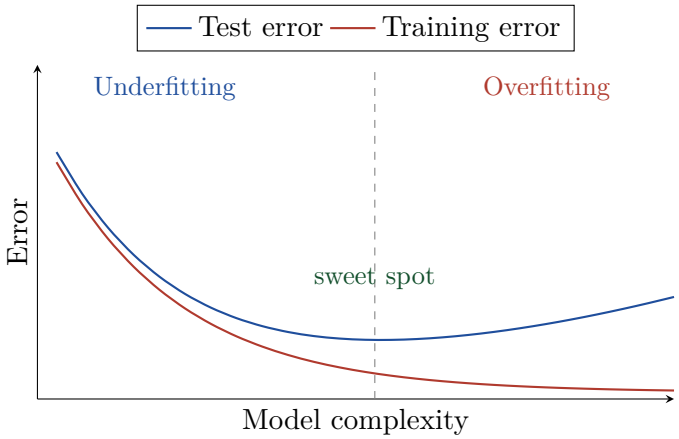
1.6.2 Problems with the algorithm

Overfitting. The model performs great on training data but poorly on new data, because it has *memorised the noise* instead of learning the pattern. It is like a student who memorises last year's paper but cannot solve a new question.

Underfitting. The model is too simple to capture the underlying pattern, so it does badly even on the training data. It is like a student who read only the chapter titles.



The classic picture below shows what happens as we make the model more complex.



Common Mistake

A very high accuracy on training data does **not** mean the model is good. Always evaluate on data the model has *never seen* (the test set, Chapter 2).

	Overfitting	**Underfitting**
Model is	too complex	too simple
Training error	very low	high
Test error	high	high
Fixes	more data, simpler model, regularisation, remove noise	more complex model, better features, train longer

Key Idea

The goal of ML is **generalisation**: performing well on *new* data, not on the data we trained with.

Chapter Summary

- AI $\supset$ ML $\supset$ Deep Learning.
- ML learns the *rules* from *data*; Mitchell's definition uses (T, P, E).
- Systems are classified as supervised, unsupervised, semi-supervised or reinforcement; batch or online; instance-based or model-based.
- Challenges: little data, non-representative data, poor quality, irrelevant features, overfitting, underfitting.
- The aim is good **generalisation** to unseen data.

Review Questions

1. Define Machine Learning. State Mitchell's definition with an example.
2. Differentiate between AI, ML and Deep Learning with a diagram.
3. Compare supervised and unsupervised learning with two examples each.
4. What is the difference between batch and online learning?
5. Distinguish instance-based and model-based learning.
6. Explain overfitting and underfitting. How can each be reduced?
7. List any four challenges of machine learning.

Practice Problems

1. Identify whether each task is classification, regression or clustering: (a) predicting tomorrow's temperature, (b) deciding if a tumour is benign, (c) grouping news articles by topic.
2. Write the (T, P, E) description for a handwriting-recognition program.
3. **Coding:** extend the `LinearRegression` example to print the predicted price for sizes 6, 7 and 8.